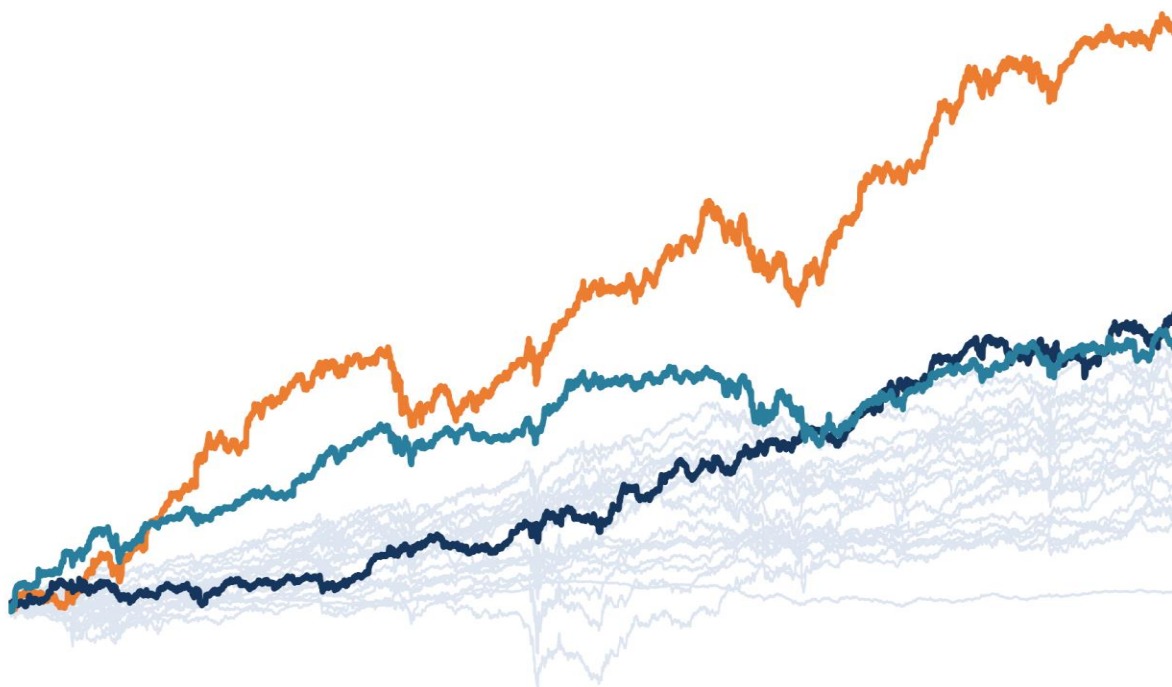


Portfolio Optimization & Risk Analytics

Backtest Reliability, Strategy Trade-offs, and Decision Support

A walk-forward historical evaluation of six portfolio construction methods across a 24-asset universe, with transaction costs, methodological auditing, factor diagnostics, and scenario-based policy analysis.



Data vintage: 2026-07-24 · frozen snapshot, SHA 8ed57bec6f01fa67

Result fingerprint: bf42baf6357816fa · identical across kernel restarts

Quality assurance: 39 of 39 automated checks passed

Executive Summary

This study asks how reliably different portfolio construction methods can be compared when the design of the backtest itself may affect the results. Before interpreting strategy performance, the backtesting engine was audited, five methodological defects were identified and corrected, and the effect of each correction was measured. Using the corrected framework, the study then compares the tested portfolio methods across historical return, risk-adjusted performance, drawdown, turnover, and transaction costs to understand both how the strategies differed and how much those conclusions depend on sound backtesting methodology.

The main takeaway

The most important result is methodological, not a claim that one strategy will win in the future. Correcting the backtest changed net CAGR by +0.41 percentage points and Sharpe by +0.05. The largest measured single step was the month-end rebalance-date correction (+0.84 pp CAGR), while allowing natural weight drift reduced CAGR by 0.40 pp. The original engine distorted results in more than one direction.

1.51 highest historical Sharpe Max Sharpe (Hist), selected on this same sample	-18.01% shallowest active drawdown Min Variance vs. -33.72% for SPY
0.39 pp largest annual cost drag Max Sharpe (Hist) at 10 bps on traded notional	Selection bias largest caveat The 24-asset universe was chosen in 2026 and contains ex-post winners

Historical performance snapshot

Strategy	Net CAGR	Volatility	Sharpe	Max DD	Ann. turnover
Max Sharpe (Hist)	28.22%	15.83%	1.51	-20.75%	1.53
Min Variance	15.53%	10.02%	1.26	-18.01%	0.50
Equal Weight	22.33%	15.70%	1.22	-28.53%	0.32
Black-Litterman	21.55%	15.13%	1.21	-28.58%	1.21
Risk Parity	18.32%	12.85%	1.19	-26.46%	0.56
HRP	15.58%	11.09%	1.15	-20.61%	0.80
SPY	15.52%	17.76%	0.77	-33.72%	0.05
60/40	9.68%	10.50%	0.71	-21.30%	0.15

How to read this table: CAGR is annualized growth; volatility is annualized variability; Sharpe is excess return per unit of volatility; maximum drawdown is the largest peak-to-trough loss; turnover measures how much the portfolio trades. These are historical descriptions of one selected universe, not forecasts.

1. Data and Evaluation Design

The analysis uses a frozen, hash-verified dataset so every reported number refers to the same market snapshot. Prices run from 2 January 2015 through 24 July 2026; the first 252 trading days are used to form the initial estimation window, leaving a common walk-forward evaluation period beginning 1 February 2016.

Item	Specification
Universe	24 securities: 22 equities across 8 sectors, 1 Treasury ETF (IEF), 1 gold ETF (GLD)
Price history	2 Jan 2015 – 24 Jul 2026 · 2,906 trading days · adjusted close
Walk-forward window	1 Feb 2016 – 24 Jul 2026 · 2,635 trading days
Rebalances	126 monthly · last trading day of each completed month
Estimation window	252 trading days ending on each rebalance date
Transaction cost	10 bps on traded notional, including initial purchase
Risk-free series	FRED DGS3MO, used point-in-time at each historical date
Reproducibility	Frozen snapshot SHA 8ed57bec6f01fa67 · result fingerprint bf42baf6357816fa

What happens at each monthly rebalance

1 · Look back	2 · Estimate	3 · Solve	4 · Trade	5 · Hold
Use the previous 252 trading days	Estimate return and covariance inputs	Compute target weights under strategy constraints	Trade from drifted pre-trade weights and pay 10 bps	Let holdings compound and weights drift until next rebalance

Estimation choices

- Covariance uses Ledoit-Wolf shrinkage, then is symmetrized and clipped to a positive semi-definite matrix.
- Historical mean returns are treated as noisy baseline estimates, not as forecasts.
- Black-Litterman uses an equal-weight equilibrium proxy because point-in-time market capitalizations are unavailable; momentum views are recomputed at each rebalance.
- Hierarchical Risk Parity uses correlation-distance clustering, quasi-diagonalization, and recursive bisection.

Why the frozen snapshot matters

A moving end date would make every rerun a different study, and adjusted-close histories can be restated after splits or dividends. Freezing the data and recording a fingerprint makes the reported results auditable and reproducible.

2. Backtest Audit: Reliability Before Results

A portfolio backtest can look mathematically sophisticated while still being wrong in small implementation details. The audit found five defects. Each correction is mechanical and testable; none depends on choosing a preferred strategy after seeing the outcome.

#	Defect	What the original engine did	Correction
1	Hidden daily rebalancing	Applied fixed target weights every day, effectively rebalancing for free	Compound asset-level holdings individually so weights drift between rebalances
2	Skipped rebalance dates	Dropped calendar month-ends that were not trading days	Use the last trading day of every completed month and audit the schedule
3	Initial cost not booked	Recorded the initial cost but divided it out of the first return	Use a gross-capital anchor so the initial cost affects net CAGR, Sharpe, and wealth
4	Risk-free look-ahead	Used a full-sample average Treasury yield at historical dates	Use the latest Treasury observation available on each rebalance date
5	Non-reproducible seeds	Derived projection seeds from Python hash(), which varies by process	Use an explicit integer seed registry and reject unregistered names

Cost and turnover conventions

Quantity	Definition	Initial purchase
Traded notional	$\sum w_{\text{target}} - w_{\text{pretrade}} $	100% - the whole portfolio is bought from cash
One-way turnover	$0.5 \times \sum w_{\text{target}} - w_{\text{pretrade}} $	50% - reported as a turnover convention
Transaction cost	Traded notional $\times$ 10 bps	10 bps of initial capital

The 0.5 factor prevents a matched sell-and-buy from being double-counted as turnover. Costs, however, are charged on the full traded notional. The initial purchase has no matching sell, so it is excluded from average ongoing turnover but still incurs the full cost.

Audit principle

A correction is not “good” because it improves performance and not “bad” because it lowers it. The standard is whether the accounting, timing, and information set match the stated methodology.

3. What Each Correction Changed

To separate the effect of each fix, one strategy is rerun through five nested implementations. Universe, estimation window, risk-free series, cost rate, and comparison window are held constant. Each row changes exactly one implementation choice from the row above it.

Variant	CAGR	Sharpe	Max DD	Turnover	Δ CAGR	Δ Sharpe
V1 · Original-like approximation	28.19%	1.48	-21.81%	1.25	—	—
V2 · + corrected month-end dates	29.02%	1.54	-20.80%	1.43	+0.84 pp	+0.062
V3 · + natural weight drift	28.62%	1.53	-20.75%	1.43	-0.40 pp	-0.007
V4 · + pre-trade turnover	28.60%	1.53	-20.75%	1.53	-0.03 pp	-0.001
V5 · Fully corrected	28.60%	1.53	-20.75%	1.53	0.00 pp	0.000

V1 is an approximation rather than a byte-exact rerun: it reuses the final notebook's estimation window and point-in-time risk-free series while reproducing the mechanical behaviors being tested.

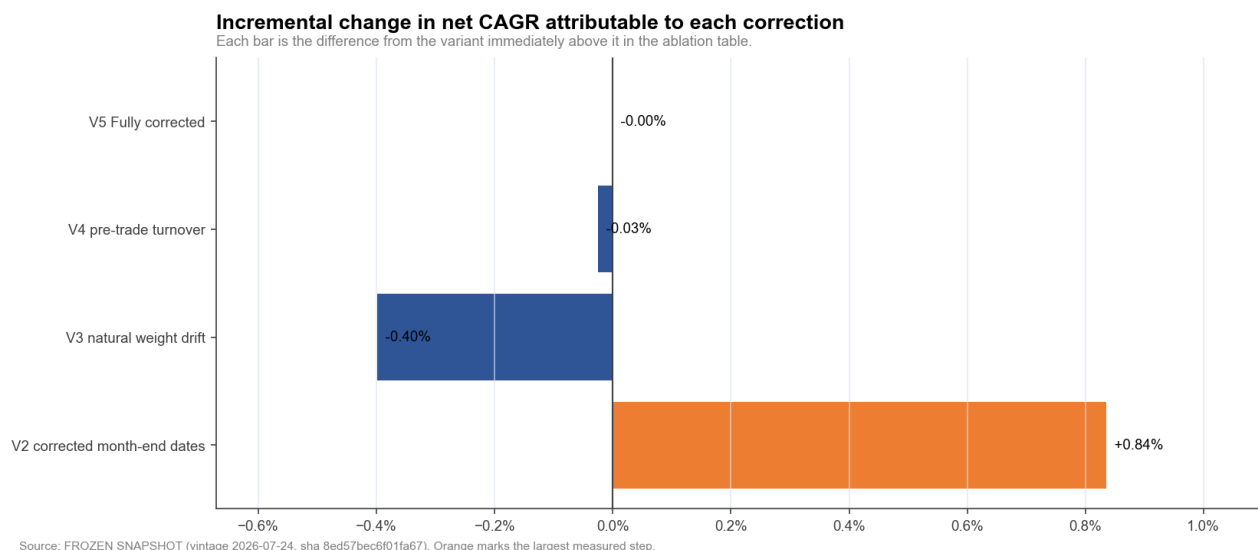


Figure 1. Incremental change in net CAGR from each correction. The month-end schedule fix had the largest measured effect.

Measured result

Moving from the original-like approximation to the fully corrected implementation raised net CAGR by 0.41 percentage points and Sharpe by 0.05. The direction was not uniform across steps, so saying “the original backtest overstated performance” would be too strong, it is more accurate to say that the original engine distorted performance.

4. Walk-Forward Historical Results

All strategies below are evaluated over the same 2,635 trading days and are shown net of transaction costs. Six portfolio construction methods are compared with three benchmarks: SPY, a 60/40 portfolio, and a diversified 60/30/10 benchmark.

Strategy	Net CAGR	Volatility	Sharpe	Max DD
Max Sharpe (Hist)	28.22%	15.83%	1.51	-20.75%
Min Variance	15.53%	10.02%	1.26	-18.01%
Equal Weight	22.33%	15.70%	1.22	-28.53%
Black-Litterman	21.55%	15.13%	1.21	-28.58%
Risk Parity	18.32%	12.85%	1.19	-26.46%
HRP	15.58%	11.09%	1.15	-20.61%
Diversified 60/30/10	10.99%	10.80%	0.81	-20.55%
SPY	15.52%	17.76%	0.77	-33.72%
60/40	9.68%	10.50%	0.71	-21.30%

Max Sharpe (Hist) had the highest return and Sharpe ratio in this sample, but it was selected using this same historical sample. Min Variance had the shallowest drawdown among the active strategies and materially lower turnover. These are descriptive results, not evidence that either ranking will persist.

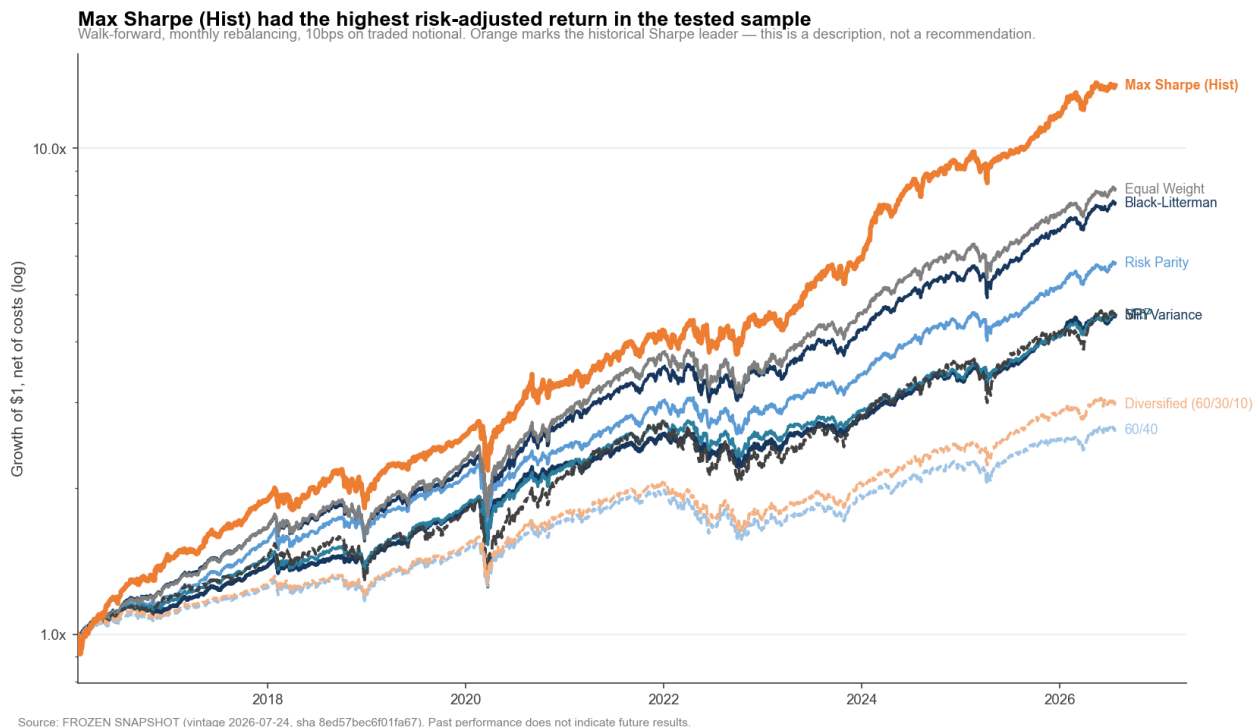


Figure 2. Cumulative growth of \$1, net of costs, on a log scale. Orange marks the historical Sharpe leader. (The chart is descriptive, not a recommendation)

5. Risk Trade-offs Behind the Returns

Return alone hides how the path felt to an investor. Two views make the trade-off clearer: drawdown shows the depth of peak-to-trough losses, while the risk-return map compares annualized return with annualized volatility.

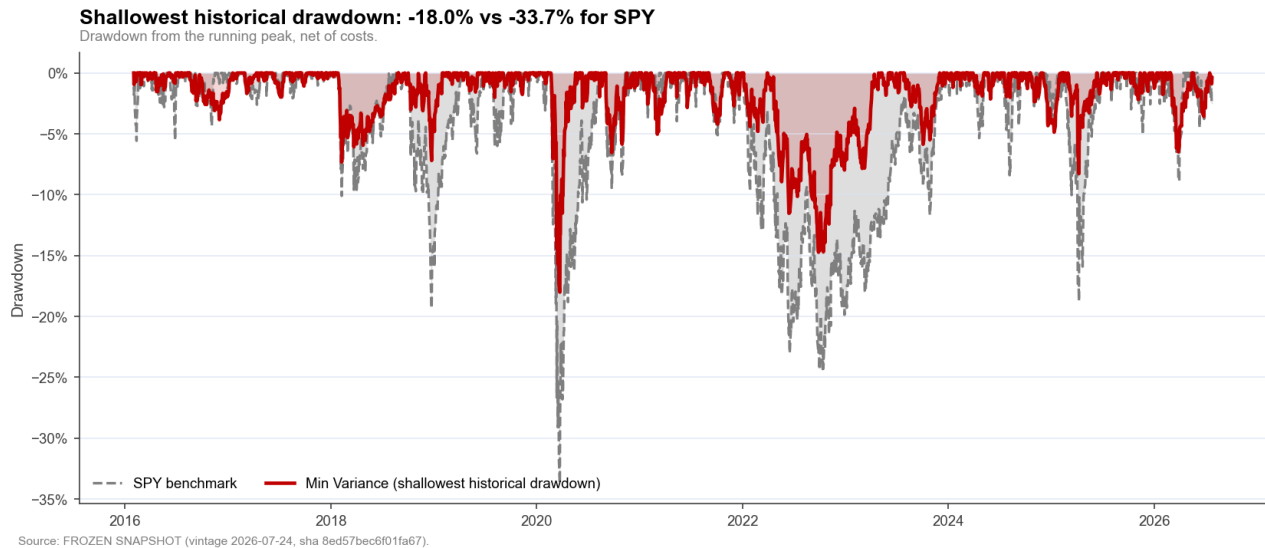


Figure 3. Drawdown from the running peak. Min Variance reached a -18.01% maximum drawdown versus -33.72% for SPY in the same window.

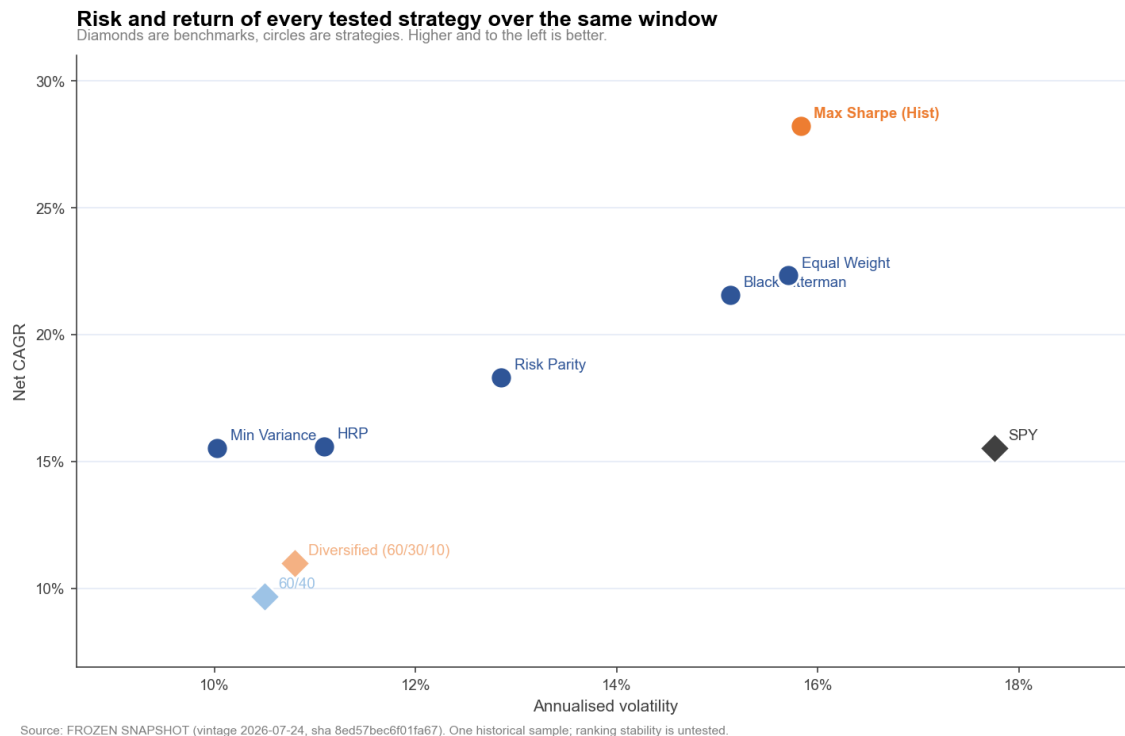


Figure 4. Risk and return across all tested strategies. Diamonds are benchmarks, circles are portfolio methods.

How to read the trade-off

The Max Sharpe strategy achieved the strongest historical return-to-volatility outcome, while Min Variance reduced the depth of losses. The chart does not establish which trade-off is preferable for a future investor, that depends on objectives and on whether these historical relationships persist.

6. Transaction Costs and Implementability

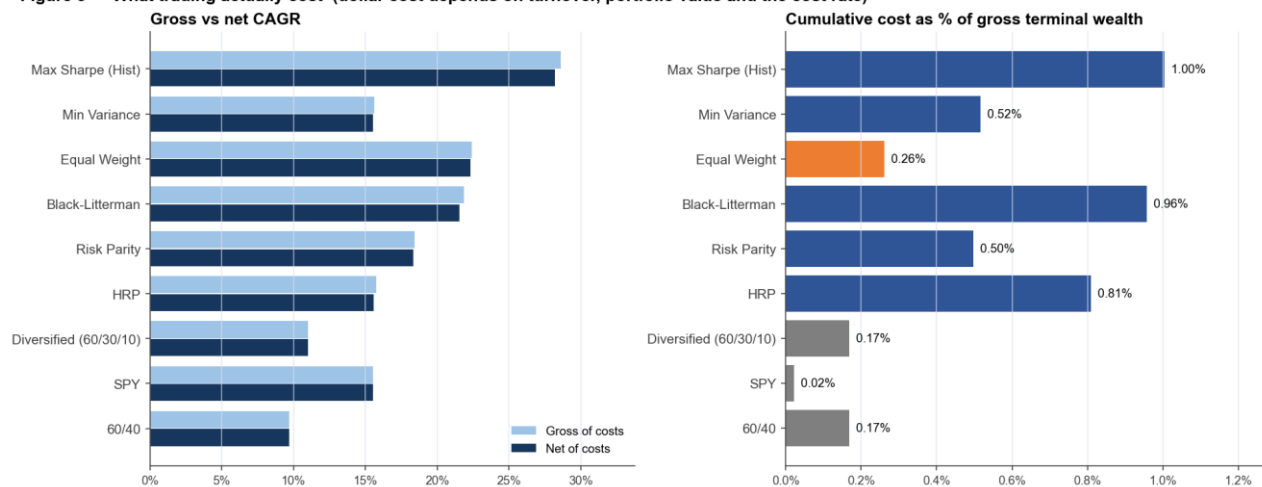
Trading costs are shown as gross-versus-net performance rather than as one headline number. This matters because the same cost can look very different depending on the denominator used.

Strategy	Gross CAGR	Net CAGR	Drag	Cost / gross terminal	Ann. turnover
Max Sharpe (Hist)	28.62%	28.22%	0.39 pp	1.00%	1.53
Black-Litterman	21.85%	21.55%	0.29 pp	0.96%	1.21
HRP	15.77%	15.58%	0.19 pp	0.81%	0.80
Risk Parity	18.45%	18.32%	0.13 pp	0.50%	0.56
Min Variance	15.64%	15.53%	0.12 pp	0.52%	0.50
Equal Weight	22.41%	22.33%	0.08 pp	0.26%	0.32
SPY	15.53%	15.52%	0.01 pp	0.02%	0.05

The same cost, three framings

For Max Sharpe, cumulative transaction costs equal \$0.1395 per \$1 of starting capital, or 13.95% of initial capital. Relative to gross terminal wealth, however, they are 1.00%, expressed as annual performance drag, they reduce CAGR by 0.39 percentage points. The first framing sounds largest partly because the denominator stays fixed while the portfolio grows.

Figure 3 — What trading actually cost (dollar cost depends on turnover, portfolio value and the cost rate)



Source: FROZEN SNAPSHOT (vintage 2026-07-24, sha 8ed57bec6f01fa67). Cost rate 10bps on traded notional. "% of initial capital" is deliberately not shown here — it flatters strategies whose wealth grew most.

Figure 5. Gross versus net CAGR and cumulative cost as a share of gross terminal wealth. Cost depends on turnover, portfolio value, and the assumed rate together.

The cost model is deliberately simple: a flat 10 bps on traded notional. It does not model bid-ask spreads by asset, market impact, taxes, or execution urgency. Cost results should therefore be read as a controlled implementation assumption rather than a complete trading-cost forecast.

7. Strategy Trade-offs: Beyond a Single Sharpe Ratio

A single metric can make a strategy look decisive when the practical trade-offs are more mixed. The scorecard ranks strategies on nine dimensions over the same historical window. The table below shows the top three by equal-weighted average rank.

Strategy	Sharpe	Max DD	Turnover	Cost drag	Eff. holdings	Stability	Avg rank
Min Variance	1.26	-18.01%	0.50	0.12 pp	9.7	0.025	2.12
Max Sharpe (Hist)	1.51	-20.75%	1.53	0.39 pp	9.7	0.115	4.00
Black-Litterman	1.21	-28.58%	1.21	0.29 pp	20.6	0.088	4.50

Stability is the average one-way turnover of the target weights themselves, so lower values indicate a steadier allocation. Effective holdings is the reciprocal Herfindahl index: higher values indicate less concentration.

Dimension	Historical leader in this sample
Return and Sharpe	Max Sharpe (Hist)
Shallowest drawdown	Min Variance
Lowest-turnover active strategy	Min Variance
Lowest cost drag	Equal Weight
Most stable target weights	Equal Weight
Most diversified by effective holdings	Equal Weight
Highest crisis-window return	Min Variance
Lowest equal-weighted scorecard rank	Min Variance

The crisis window is defined mechanically as SPY's deepest drawdown in the sample: 19 February to 23 March 2020, when SPY fell 33.7%. Min Variance lost 17.88% over that window. Equal weighting across scorecard dimensions is transparent but still a modeling choice, different priorities can produce a different ordering.

8. Factor Analysis: Interpreting “Alpha” Carefully

Two diagnostic portfolios are regressed on the Fama-French five factors using Newey-West (HAC) standard errors with 21 lags. The objective is to describe what the realized return series is related to, not to certify investment skill.

Portfolio	Alpha (ann.)	t-stat	p-value	Market beta	R ²
Max Sharpe (Hist)	15.60%	6.59	0.000	0.85	0.78
Equal Weight	6.99%	6.27	0.000	0.87	0.94

Factor data end on 29 May 2026, 56 days before the price series, because the Kenneth French data are published with a lag. Regressions therefore use only the overlapping period.

Why a statistically significant intercept is not proof of skill

First, the Max Sharpe portfolio was selected using the same sample that the regression evaluates, so the intercept is upward-biased by selection. Second, the five-factor specification omits other possible exposures such as momentum, quality, and low volatility. Third, statistical significance in one historical sample does not establish persistence.

For readability, the table above emphasizes the quantities most useful to a non-specialist reader. The notebook retains the complete factor loadings (SMB, HML, RMW, and CMA) and the full regression output.

9. Investment Policy Engine: Illustrative Allocations

The policy engine translates three risk profiles into constrained allocations using Black-Litterman model inputs. This section is intentionally separated from the walk-forward evidence: the policy weights are calibrated on the full sample, so they are decision-support illustrations rather than out-of-sample performance results.

Profile	Model-implied return	Volatility	Eff. holdings	Equity	Fixed income	Commodity
Conservative	10.91%	9.33%	7.5	60.0%	20.0%	20.0%
Balanced	14.84%	11.82%	15.2	71.3%	14.3%	14.5%
Aggressive	18.93%	15.79%	23.1	91.7%	4.2%	4.2%

The asset taxonomy keeps three concepts separate: asset class, sector, and portfolio role. For example, a defensive utility stock remains an equity even if it behaves as ballast. This prevents “defensive” exposure from being mistaken for genuine non-equity diversification.

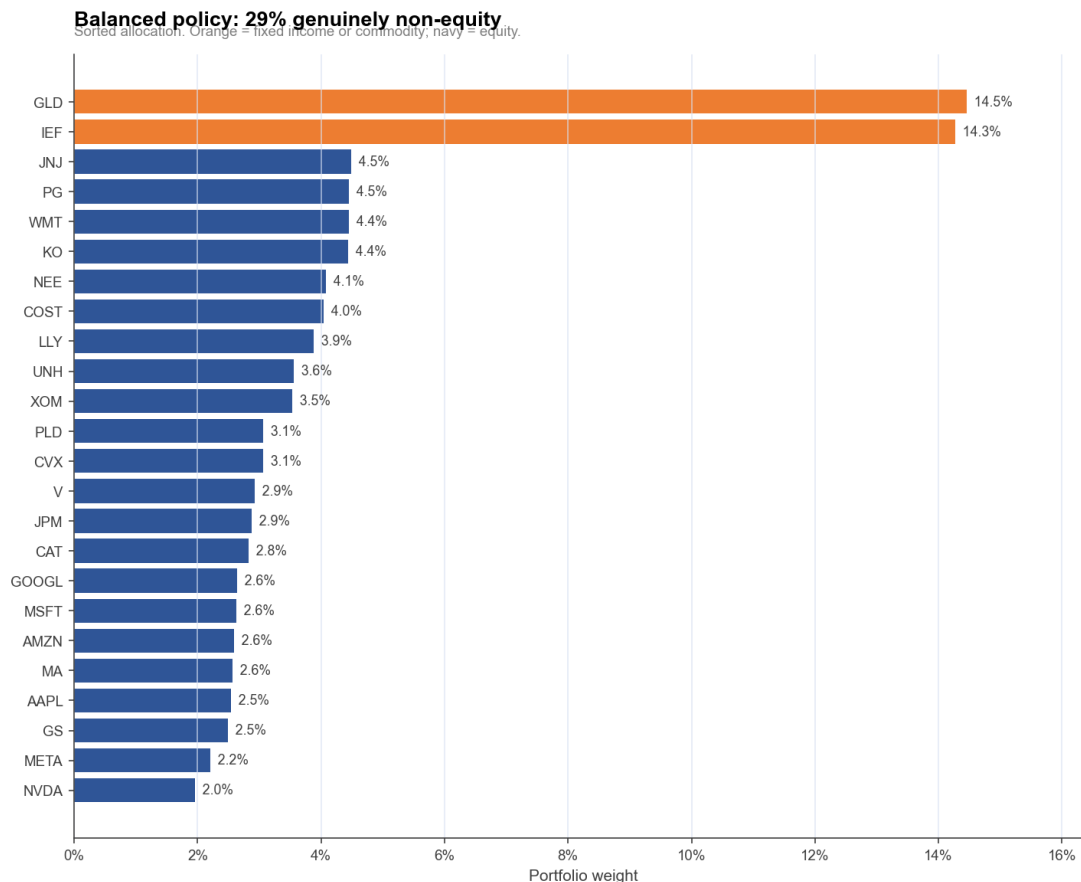


Figure 6. Balanced policy allocation. Orange marks the genuinely non-equity holdings (Treasury and gold ETFs); blue bars are equities.

Interpretation

The policy engine is useful for translating risk preferences into a consistent set of constraints and exposures. Its model-implied returns should not be read as expected realized returns, because the inputs are calibrated on this historical sample and inherit the same selection bias.

10. Forward Scenario Simulation – Not a Forecast

Read this section as a model experiment

Each path is produced by resampling blocks from the historical 24-asset return matrix. Percentiles describe the simulated distribution under those assumptions only. A 5th-percentile outcome is not a floor, and a high simulated frequency is not a guarantee. Selection bias in the historical universe carries directly into the simulations.

Method: multivariate moving-block bootstrap. Asset-return blocks are resampled together to preserve cross-sectional correlation and volatility clustering. The policy allocation is then applied to each path, with rebalancing every 21 trading days and transaction costs charged inside the simulation. The reported experiment uses 4,000 paths, a 10-year horizon, and \$100,000 of initial capital.

Profile	Median value	5th pct	95th pct	Median CAGR	Paths $\geq 2\times$	Median sim. Max DD
Conservative	\$291,565	\$183,143	\$455,637	11.29%	3,637 / 4,000	-14.5%
Balanced	\$460,804	\$269,261	\$780,829	16.51%	3,980 / 4,000	-18.7%
Aggressive	\$706,455	\$337,254	\$1,394,245	21.59%	3,989 / 4,000	-24.4%

Raw path counts are shown so rounded percentages are not mistaken for certainty. For example, the Balanced profile doubled in 3,980 of 4,000 simulated paths (99.5%), not in every path. Changing the block length from 21 to 42 or 63 days moved median outcomes by only 0.6%–1.8% across profiles.

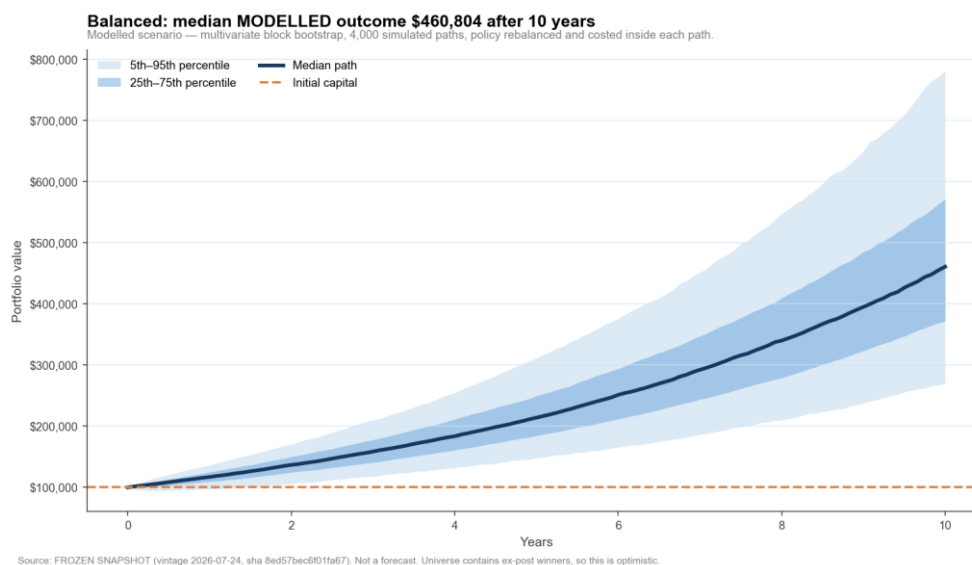


Figure 7. Balanced profile: simulated distribution of portfolio value over ten years. Bands are model percentiles, not confidence intervals for the true future.

11. Quality Assurance

Thirty-nine automated checks run at the end of every execution. The pass banner is conditional: failures print the failing rows rather than a blanket confirmation. Every optimization in the reported run converged and passed weight validation; no fallback portfolio was required across 756 solves.

Category	Checks	Representative evidence
Data integrity	4	0 NaNs; no duplicate dates; identical comparison window
Risk-free point-in-time	4	756 date pairs verified; historical rate lookup audited
Rebalance schedule	5	126 valid dates; 0 empty holding periods; incomplete final month excluded
Look-ahead	2	Estimation ends by rebalance; holding starts strictly after
Weights & constraints	6	0 / 504 bound violations; max budget deviation 4.44e-16
Transaction costs	7	Cost reconciliation error 2.78e-17; higher cost rates never improve net results
Reproducibility	2	Identical draws across runs; 10 explicit seeds
Terminology & labels	3	No automatic “recommended” label; projections labeled as modelled
Snapshot integrity	5	SHA verified; CSV round-trip bitwise exact
Determinism	1	Result fingerprint bf42baf6357816fa

What QA can and cannot do

Passing tests increases confidence that the code implements the stated methodology consistently. It does not remove economic model risk, selection bias, or uncertainty about whether relationships observed from 2015-2026 will hold in a different market regime.

12. Limitations and Responsible Interpretation

Primary limitation: survivorship and selection bias

The 24-asset universe was chosen in 2026 and includes companies that are prominent partly because they succeeded. The dataset does not contain point-in-time index membership, so this cannot be repaired inside the current study. Every backtest return and every scenario percentile is therefore optimistic by an amount that cannot be quantified here.

Other limitations

- Black-Litterman uses an equal-weight equilibrium proxy rather than the market portfolio. The momentum view, 2% spread, $\tau = 0.05$, and Ω construction are analyst choices rather than measured certainty.
- Forward simulations hold the policy weights fixed inside each path rather than re-optimizing, and they cannot generate regimes that are absent from the 2015-2026 history.
- Transaction costs are linear in traded notional. Market impact, taxes, liquidity differences, and execution urgency are excluded.

- Policy-engine weights are calibrated in-sample. The walk-forward strategy evaluation is the section that provides the cleanest historical evidence.
- The study observes one 2015-2026 market path, which was generally favorable to many of the selected assets. Strategy ordering is not shown to be stable across independent regimes.
- Reproducibility is exact within the fixed environment, different scipy/scikit-learn versions can produce slightly different SLSQP solutions.

13. Conclusions

1. Min Variance produced the shallowest historical drawdown among the tested active strategies (-18.01% versus -33.72% for SPY) while trading roughly one-third as much as the highest-turnover method.
2. Max Sharpe (Hist) recorded the highest historical Sharpe ratio (1.51) and net CAGR (28.22%), but it was selected on the same sample in which it is evaluated and therefore carries selection and estimation risk.
3. Transaction costs mattered most for high-trading strategies: the largest annual CAGR drag was 0.39 percentage points, compared with 0.08 for Equal Weight and 0.01 for SPY under the same 10 bps assumption.
4. The backtest audit changed net CAGR by 0.41 percentage points, which is large enough to exceed the gaps between several competing strategies. Therefore, backtest construction itself is a material source of model risk.